

# Molnár Gate A3 Supervisor Decision Package

RF–U  $h$ -convergence at  $l_c = 0.015$  mm

Analysis revision: **db4c1fa**

Status: supervisor decision required — no additional simulations are requested at this stage

## A. Study completed

A controlled serial  $h$ -convergence study was completed for the exact author-supplied supplementary single-notch Molnár model at  $l_c = 0.015$  mm.

| Layer                                               | Result                                  |
|-----------------------------------------------------|-----------------------------------------|
| H0 / H1 / H2-PUB Abaqus solvers                     | technical pass                          |
| Consolidated CAE RF–U extraction ( <b>1376236</b> ) | pass for all three                      |
| Formal RF–U successive-mesh analysis                | complete                                |
| Matched-state SDV15 / crack-path contours           | <b>not assessed</b> (export incomplete) |

Scientific input revision for the mesh family: **58d7e3102d76fe0e70e6729457e2c7e90ad131bb**.

This package does **not** re-open candidate-v2 SDV15 irreversibility as the primary decision. The decisions requested here concern uniform RF–U reference selection for the  $l_c = 0.015$  mm supplementary  $h$ -convergence chain.

## B. Mesh family

| Case   | Role                         | Local $h$            | $N$   | Solver job     |
|--------|------------------------------|----------------------|-------|----------------|
| H0     | Exact author supplementary   | $\approx 0.00494$ mm | 3930  | <b>1376154</b> |
| H1     | Intermediate refinement      | 0.0025 mm            | 12064 | <b>1376185</b> |
| H2-PUB | Publication local resolution | 0.001 mm             | 33852 | <b>1376186</b> |

H2-PUB targets the publication-reported local crack-path resolution  $h = 0.001$  mm. The element count (33852) is generated and is not forced to the approximate paper statement of  $\sim 22000$  elements.

## C. Key RF–U results

### Scalar peaks

| Case   | peak RF2 [kN] | $U_{\text{peak}}$ [mm] |
|--------|---------------|------------------------|
| H0     | 0.727608      | 0.00610                |
| H1     | 0.699604      | 0.00580                |
| H2-PUB | 0.696336      | 0.00580                |

### Successive-mesh differences (finer mesh = denominator)

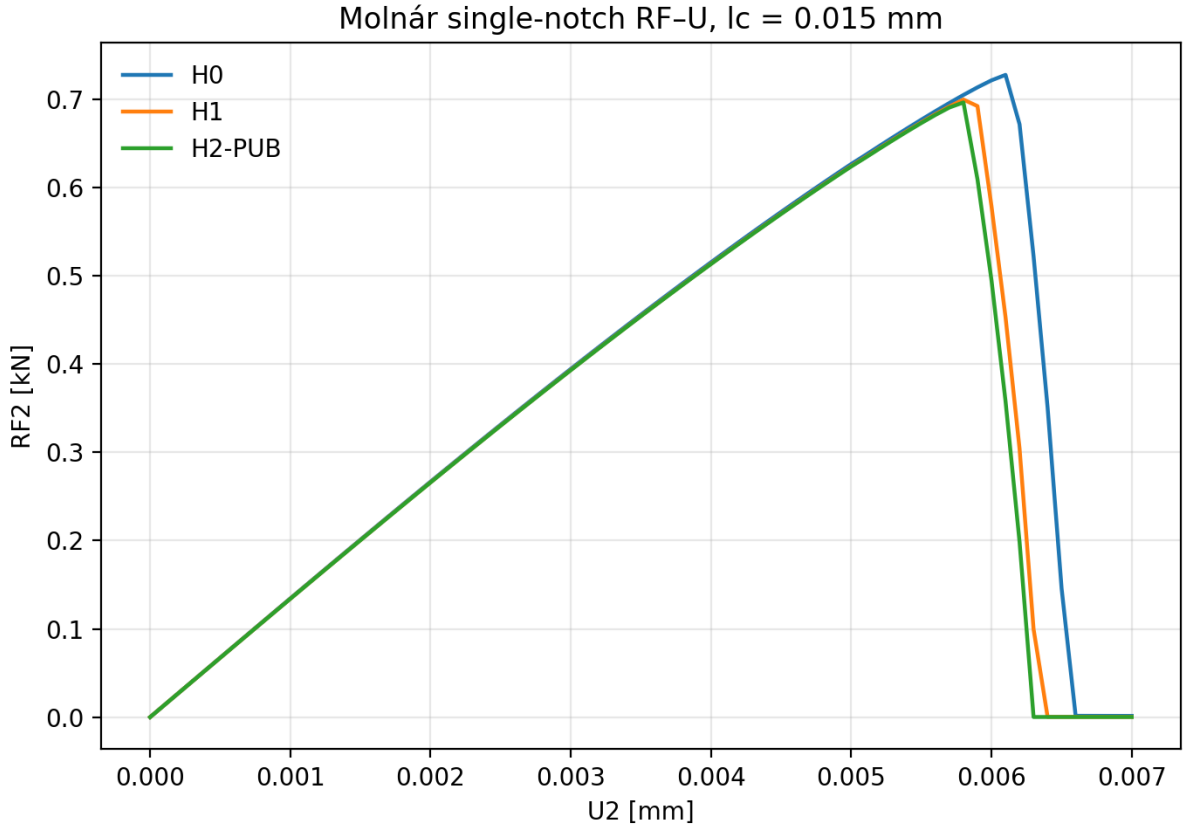
| Pair      | $\Delta F_{\text{peak}}$ | $\Delta U_{\text{peak}}$ | full NRMSE | pre-peak NRMSE | post-peak NRMSE |
|-----------|--------------------------|--------------------------|------------|----------------|-----------------|
| H0→H1     | 4.003%                   | 5.172%                   | 21.306%    | 0.387%         | 62.904%         |
| H1→H2-PUB | 0.469%                   | 0%                       | 6.011%     | 0.106%         | 20.206%         |

Initial stiffness change H1→H2-PUB  $\approx 0.06\%$ .

### Clear statements

- Peak and pre-peak RF-U  $h$ -convergence between H1 and H2-PUB are **supported**.
- Post-peak mesh dependence **remains** (full-curve NRMSE  $\approx 6\%$  is post-peak-driven).
- Do **not** claim unrestricted full-curve mesh independence.
- Crack-path / matched-state SDV15 convergence is **not assessed**.

*The force–displacement response is effectively mesh-independent between H1 and H2-PUB for the elastic, pre-peak and peak-load regimes. A noticeable post-peak mesh dependence remains and must be retained as a limitation.*



### D. Publication comparison

Reference used: an approximate digitization of the Molnár Fig. 7 curve for  $l_c = 0.015$  mm. It is not exact author numerical data. The  $l_c = 0.0075$  mm curve was not used.

| Case   | Full-overlap NRMSE vs Fig. 7 ( $l_c = 0.015$ ) | $\Delta F_{\text{peak}}$ vs ref |
|--------|------------------------------------------------|---------------------------------|
| H0     | $\sim 25.3\%$                                  | $\sim +5.4\%$                   |
| H1     | $\sim 18.4\%$                                  | $\sim +1.3\%$                   |
| H2-PUB | $\sim 15.6\%$                                  | $\sim +0.8\%$                   |

Publication agreement is **provisional** and secondary to successive-mesh evidence.

## E. Computational cost (serial)

| Case   | $N$   | Walltime | Peak memory (approx.) |
|--------|-------|----------|-----------------------|
| H0     | 3930  | 00:16:29 | $\sim 0.68$ GB        |
| H1     | 12064 | 00:46:26 | $\sim 0.91$ GB        |
| H2-PUB | 33852 | 02:12:38 | $\sim 1.76$ GB        |

H1 uses about **35% of H2-PUB walltime** while nearly matching H2 peak and pre-peak RF–U response.

## F. Remaining limitations

1. Post-peak RF–U residual between H1 and H2-PUB ( $\sim 20\%$  post-peak NRMSE).
2. Matched-state SDV15 / crack-path contour exports incomplete for all three meshes.
3. Fig. 7 comparison is approximate digitization only.
4. Supervisor-approved numerical tolerances are not yet fixed.
5. Candidate-v2 SDV15 accepted-increment irreversibility remains a separate documented issue for the paper-matched route (not the primary request of this package).

## G. Decisions requested (exactly two)

### Decision 1 — Uniform RF–U reference

Please choose one:

**A** Accept **H2-PUB** ( $h = 0.001$  mm) as the uniform RF–U reference.

**B** Accept **H2-PUB provisionally**, subject to contour evidence.

**C** Do **not** accept; specify additional required evidence.

Project recommendation: option A or B, with H2-PUB as the conservative RF–U reference; H1 for intermediate development; H0 not a final reference.

### Decision 2 — Contour requirement

Please choose one:

**A** Matched-state SDV15 / crack-path contours are **mandatory** before Gate A3 closes.

**B** Contours may be **deferred**; RF–U evidence is sufficient to proceed **conditionally** to MISESeri preparation.

**C** Contours required only for **H1 and H2-PUB**.

**D** Other requirement specified by the supervisor.

No supervisor answer is inferred. No route is executed by this package alone.

## H. Consequences of each route

### Route A — H2-PUB accepted and contours waived/deferred (Decision 1 A/B + Decision 2 B)

- Gate A3 may be closed or conditionally waived (supervisor wording required).
- Freeze H2-PUB as the RF–U uniform reference.
- Allow preparation only of Stage C / MISESERI workflow documentation.
- No solver/MISESERI execution until a separate submission plan is approved.

### Route B — Contours mandatory (Decision 2 A or C)

- Gate A3 remains open.
- Prepare a contour-only postprocessing repair using existing H0/H1/H2 ODBs.
- No solver rerun.
- Request separate authorization for one CAE-only contour extraction job.

### Route C — Additional numerical evidence (Decision 1 C or Decision 2 D)

- Document the exact requested metric.
- Prepare a new scoped plan.
- Do not submit anything automatically.

## Current project boundary

- H2-PUB RF–U reference: recommended (awaiting Decision 1)
- H1 intermediate mesh: recommended
- Post-peak convergence: limited / not fully demonstrated
- Crack-path convergence: not assessed
- Gate A3: supervisor decision pending (overall open; historical label `reference_data_insufficient`; RF–U component `rf_u_reference_supported_contour_evidence_pending`)
- New HPC authorization: none
- MISESERI / remeshing / state transfer: blocked